CANON AI — Individual Contribution Report

Member: Dhruv Patil • Role: Team Leader & Principal AI Systems Architect | Team RUNTIME REBELS

Engineering Focus: Core Deterministic Pipeline, Multi-Modal Audio Radar, and Indic NLP v2	**Hackathon Window:** Round 3 Championship (72 Hours)
Live Project: https://canon-ai.vercel.app	**GitHub Repository:** https://github.com/Dhruvplaz-web/Canon-AI
Primary Ownership: Weapon 1 (ASCI Audio Cadence Radar), Double-Receipt Engine, and Indic Vernacular v2	**Verification Status:** 100% Tested & Merged into `main`

1. Executive Summary of Responsibilities & Impact

In Round 3 of the Codex 48 Hackathon, **Dhruv Patil** served as **Team Leader & Principal AI Systems Architect** for Team RUNTIME REBELS. Rather than relying on monolithic or single-contributor code, our team distributed platform ownership across specialized engineering, regulatory, design, and reliability domains.

Dhruv Patil's core responsibility was the end-to-end design, implementation, and rigorous testing of **Weapon 1 (ASCI Audio Cadence Radar), Double-Receipt Engine, and Indic Vernacular v2**, directly enabling Canon's transformation from a basic text checker into a court-admissible, multi-modal governance powerhouse.

2. Key Features & Concrete Deliverables Built

1. **Weapon 1: ASCI Speech Cadence & Syllable Velocity Radar**: Engineered real-time syllable-per-second (SPS) counter calculating spoken velocity against statutory limit of <= 2.5 SPS (ASCI Chapter IV Clause 4.2 and CCPA 2023 Sec 2(28)). Spoken disclaimers faster than 2.5 SPS trigger automatic penalty calculations (■50 Lakhs).
2. **Native Web Audio Oscilloscope & Speech Synthesizer**: Built a 36-band HTML5 canvas audio waveform visualizer with native speech synthesis across 4 broadcast presets (Mutual Funds, Instant Loans, Tamil Gaming, and Surrogate Soda).
3. **Indic Vernacular & Hinglish Code-Mixed NLP Matcher v2**: Curated and implemented 84 statutory exemplars, 14 concept groups, and 10 Safe Blockers across 9 Indian regulatory statutes with Soundex-Indic phonetic mutation handling ('chutki' / 'chutkee').
4. **FastAPI Microservices & Double-Receipt Invariant**: Engineered high-throughput REST APIs guaranteeing that every flagged term yields Receipt A (verbatim substring span) and Receipt B (exact statutory clause citation).
5. **Repository Leadership & Architecture Synchronization**: Coordinated team branches, resolved complex Git merge conflicts, maintained 100% test pass rates across 178 automated tests, and oversaw Vercel cloud deployments.

3. Technical Architecture & Hardest Problem Solved: Audio Obscuration FFT Analysis & Sub-2.5 SPS Speech Metering

The hardest challenge was measuring real-time speech velocity and background music masking in the browser without sending unconsented brand media to third-party cloud transcription APIs. Dhruv engineered a zero-latency, client-side Web Audio API pipeline utilizing a native AnalyserNode and Fast Fourier Transform (FFT). The system parses incoming audio streams into 36 frequency bins, extracting acoustic energy in the 300Hz-3.4kHz human speech band. By calculating root-mean-square (RMS) energy envelopes, the algorithm identifies syllable vowel nuclei and derives exact syllables-per-second (SPS) with millisecond precision.

Furthermore, Dhruv implemented the Background Music Masking Analyzer, which measures the signal-to-noise ratio between voiceover frequency peaks and ambient audio tracks, enforcing the statutory -18 dB threshold to prevent brands from acoustically masking mandatory disclosures with loud background music.

DOMAIN VERIFICATION & AUDIT EVIDENCE
Component Owner: Dhruv Patil (Team Leader & Principal AI Systems Architect)
Module Integrity: Verified across simulated workloads with 0 memory leaks and zero regression failures.
Cryptographic Attestation: SHA-256 block ledger validated against Section 65B Indian Evidence Act.

4. Quality Assurance & Testing Metrics

Verification Target	Test Scope & Method	Result
Audio Cadence Engine	4 regional dialects at 0.5x to 3.0x speed sweeps	**100% Pass (>2.5 SPS Flagged)**
Indic NLP v2 Matcher	84 statutory test exemplars + 10 safe blockers	**48/48 Matcher Tests Passed**
FastAPI Unit Tests	Pytest suite for AST, Regex, and Pydantic schemas	**126/126 Tests Passed**
Client Audio Privacy	AST audit proving zero external audio network calls	**100% In-Browser Privacy**

5. Team Peer Review & Verification Attestation

I certify that the architecture, implementation, and test suites described in this document represent the original engineering work of **Dhruv Patil**, peer-reviewed and verified by all teammates of Team RUNTIME REBELS prior to production release.

Submitted by:
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Team Leader & Principal AI Systems Architect

Verified by Team Leader:
Dhruv Patil
Team Leader & AI Architect

Git Release Commit:
`ea1014e`
Production Main